

Knowledge Graph Link Prediction with Deep Knowledge Graph Embedding Models

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Deep Learning & Knowledge Graphs — unified project

Problem and motivation. A *knowledge graph* collects facts as triples (h, r, t) : head entity h , relation r , tail entity t . *Link prediction* answers queries $(h, r, ?)$ or $(?, r, t)$: given the training facts, which missing endpoint is most plausible? We follow standard knowledge graph embedding [2]: learn a score $f_\theta(h, r, t) \in \mathbb{R}$ so true triples rank above corrupted ones.

Pipeline (build the KG, then predict). The project matches both courses in order: **(1) Dataset $\rightarrow$ graph.** We use the public **FB15k-237** benchmark [3]—released as **real triple files** (train/validation/test splits). That release *is* our dataset: thousands of facts about real-world entities, not a toy hand-drawn sketch. **Building** the KG means ingesting those triples, mapping strings to entity/relation IDs, forming the edge set $\mathcal{T}$ and the split, and loading the result into PyKEEN [1]; we may also export a small RDF or edge-list slice for figures or the KG report. **(2) Prediction.** We then train models only on $\mathcal{T}_{\text{train}}$ and evaluate on held-out test triples (next paragraph). We do not invent a private ontology from scratch, but we still **materialize** the full multi-relational graph the methods run on.

What the graph looks like. Nodes are entities; each fact is a **directed, labeled** link $h \xrightarrow{r} t$. Many relation types exist, and the same pair (h, t) can appear under different r . Equivalently, $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ with $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. We keep the official benchmark partition unchanged (no extra subsampling).

Task and evaluation protocol. Training uses only $\mathcal{T}_{\text{train}}$: we minimize a **margin/ranking loss** and draw **negative** triples by corrupting head or tail (standard pipeline [2, 3]). At test time, each $(h, r, t) \in \mathcal{T}_{\text{test}}$ was unseen during training; we score all candidate completions and record the rank of the true entity. **Filtered** metrics [2]: when ranking the true tail for $(h, r, ?)$, any other tail t' with $(h, r, t') \in \mathcal{T}_{\text{train}} \cup \mathcal{T}_{\text{val}} \cup \mathcal{T}_{\text{test}}$ is treated as ambiguous (not penalized as an error); the same for head prediction. We average results over head and tail prediction.

Scale. FB15k-237 (Freebase-derived) counts: **14,541** entities, **237** relations, **272,115** train / **17,535** validation / **20,466** test triples—large enough to be meaningful, still feasible on Colab/GPU. An optional second graph (e.g. OGB link prediction) is possible; FB15k-237 remains the main result.

Method (planned). **Baseline:** TRANSE [2]. **Stronger model:** ROTATE [3]. Both via PyKEEN [1] (same splits, negative sampling, and evaluator for fair comparison). We will record embedding dimension d , loss, optimizer, batch size, number of epochs (or early stopping on validation MRR), learning rate, seed, and GPU/CPU time. Course material fills in background reading.

Metrics and qualitative analysis. **Quantitative:** mean reciprocal rank (MRR) and Hits@1, Hits@3, Hits@10 under the filtered protocol, reported for head and tail (then averaged, as is standard). **Qualitative:** loss and validation MRR (or Hits@10) vs. epoch, plus a short look at a few test queries and their top-ranked entities.

References

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